Exercises: Work, energy and power

IBDP Physics SL · Grade 11 · A.3 Work and energy · 24 calculation exercises in three IB levels

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Name: Class: Date:

How to use. Show your working: equation → substitution with units → answer with unit and sensible significant figures. Use g = 9.81 m s^{-2}. Level 1 = one step (Paper 1 style), Level 2 = multi-step, Level 3 = structured exam questions with marks.
Check yourself online (instant marking, hints and worked solutions, new numbers to practise again):
https://school.taneski.com/physics/dp/ex-work-energy-power/

Level 1 · Paper 1 style — one step

8 exercises · 14 marks · show your working

1.1 Lifting a box

A.3 · Work and energy · [2 marks]

A 3.0 kg box is lifted vertically through 1.7 m at constant speed. Calculate the work done on the box by the lifting force.

Answer: _____ J

1.2 Pulling at an angle

A.3 · Work and energy · [2 marks]

A force of 40 N acting at 50° above the horizontal pulls a box 14 m along a level floor. Calculate the work done by the force.

Answer: _____ J

1.3 Runner's kinetic energy

A.3 · Work and energy · [1 mark]

Calculate the kinetic energy of a 72 kg runner moving at 10 m s^{-1}.

Answer: _____ J

1.4 Speed from kinetic energy

A.3 · Work and energy · [2 marks]

A 0.43 kg football has 60 J of kinetic energy. Calculate its speed.

Answer: _____ m s^{-1}

1.5 Stretched spring

A.3 · Work and energy · [2 marks]

A spring with spring constant 100 N m^{-1} is stretched by 6.0 cm. Calculate the elastic potential energy stored.

Answer: _____ J

1.6 Winch

A.3 · Power and efficiency · [2 marks]

A winch lifts a 22 kg load through 4.0 m at constant speed in 5.0 s. Calculate the useful power output of the winch.

Answer: _____ W

1.7 Motor efficiency

A.3 · Power and efficiency · [1 mark]

An electric motor takes in 220 W and delivers 170 W of useful mechanical power. Calculate its efficiency.

Answer: _____ %

1.8 Hair dryer A.3 · Power and efficiency · [2 marks]

A 2.0 kW hair dryer is used for 12 minutes.

(a) Calculate the energy used in kWh. [1]

(a) Answer: _____ kWh

(b) Calculate the energy used in joules. [1]

(b) Answer: _____ J

Level 2 · Multi-step — two or three linked steps

8 exercises · 31 marks · show your working

2.1 Swinging pendulum A.3 · Energy conservation · [4 marks]

A pendulum of length 1.2 m is pulled aside until the string makes 35° with the vertical, and is released from rest. Air resistance is negligible.

(a) Calculate the height through which the bob falls to its lowest point. [2]

(a) Answer: _____ m

(b) Calculate the maximum speed of the bob. [2]

(b) Answer: _____ m s^{-1}

2.2 Braking distance from work done A.3 · Work and energy · [3 marks]

A 1700 kg car travelling at 32 m s^{-1} brakes with a constant braking force of 7.0 kN until it stops. Use the work–energy principle to calculate the braking distance.

Answer: _____ m

2.3 Pushing against friction A.3 · Work and energy · [3 marks]

A 3.0 kg box at rest is pushed 4.0 m across a floor by a horizontal force of 22 N. The friction force is 9.0 N.

(a) Calculate the net work done on the box. [2]

(a) Answer: _____ J

(b) Calculate the final speed of the box. [1]

(b) Answer: _____ m s^{-1}

2.4 Cart hits a spring A.3 · Energy conservation · [4 marks]

A 0.60 kg cart moving at 2.2 m s^{-1} on a smooth track runs into a spring of spring constant 240 N m^{-1}.

(a) Calculate the maximum compression of the spring. [3]

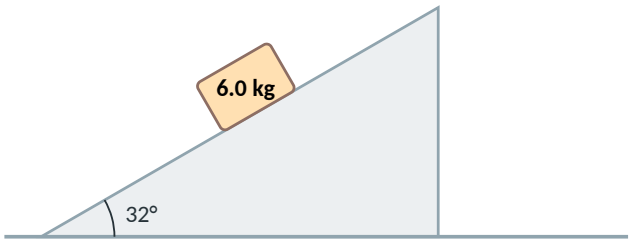
(a) Answer: _____ m

(b) Calculate the maximum force exerted by the spring on the cart. [1]

(b) Answer: _____ N

2.5 Block down a rough slope A.3 · Work and energy · [5 marks]

A 6.0 kg block slides 4.0 m from rest down a slope inclined at 32°. The coefficient of dynamic friction is 0.22.



(a) Calculate the work done by gravity. [1]

(a) Answer: _____ J

(b) Calculate the work done against friction. [2]

(b) Answer: _____ J

(c) Calculate the speed of the block after 4.0 m. [2]

(c) Answer: _____ $m\ s^{-1}$

2.6 Trolleys stick together A.3 · Energy conservation · [4 marks]

A 3.0 kg trolley moving at 4.0 $m\ s^{-1}$ collides with a 2.0 kg trolley at rest. They stick together.

(a) Calculate the common velocity after the collision. [1]

(a) Answer: _____ $m\ s^{-1}$

(b) Calculate the kinetic energy lost in the collision. [2]

(b) Answer: _____ J

(c) Calculate the percentage of the kinetic energy lost. [1]

(c) Answer: _____ %

2.7 Water pump A.3 · Power and efficiency · [4 marks]

A pump raises 340 litres of water per minute through a height of 14 m. The pump is 80 % efficient. (1 litre of water has a mass of 1 kg.)

(a) Calculate the useful power output of the pump. [2]

(a) Answer: _____ W

(b) Calculate the input power of the pump. [2]

(b) Answer: _____ W

2.8 Cyclist's power A.3 · Power and efficiency · [4 marks]

A cyclist rides at a constant 9.0 m s^{-1} against total resistive forces of 27 N.

(a) Calculate the useful power output of the cyclist. [1]

(a) Answer: _____ W

(b) Calculate the useful energy transferred in 40 minutes, in kJ. [1]

(b) Answer: _____ kJ

(c) The cyclist's muscles are 25 % efficient. Calculate the food energy used in this time, in kJ. [2]

(c) Answer: _____ kJ

Level 3 · Paper 2 structured — parts and marks

8 exercises · 60 marks · show your working

3.1 Roller coaster with friction A.3 · Energy conservation · [8 marks]

A 650 kg roller-coaster car starts from rest at the top of a drop 45 m above the bottom. The track along the drop is 150 m long. The car reaches the bottom at 28 m s^{-1}.

(a) Calculate the gravitational potential energy lost by the car. [1]

(a) Answer: _____ J

(b) Calculate the kinetic energy of the car at the bottom. [1]

(b) Answer: _____ J

(c) Calculate the average resistive force on the car along the track. [3]

(c) Answer: _____ N

(d) Calculate the efficiency of the transfer from gravitational potential energy to kinetic energy. [2]

(d) Answer: _____ %

(e) Calculate the speed the car would have at the bottom without any resistance. [1]

(e) Answer: _____ m s^{-1}

3.2 Catapult A.3 · Energy conservation · [7 marks]

A catapult has a spring constant of 500 N m^{-1}. It is stretched by 0.18 m and fires a 40 g stone vertically upwards.

(a) Calculate the elastic potential energy stored in the catapult. [1]

(a) Answer: _____ J

(b) Calculate the launch speed of the stone (assume all the elastic energy becomes kinetic energy). [2]

(b) Answer: _____ m s^{-1}

(c) Calculate the maximum height the stone would reach without air resistance. [2]

(c) Answer: _____ m

(d) The stone actually rises 16.0 m. Calculate the average air-resistance force on the way up. [2]

(d) Answer: _____ N

3.3 Small hydroelectric scheme

A.3 · Power and efficiency · [8 marks]

Water flows at 250 kg s^{-1} from a lake through a turbine 40 m below the lake's surface. The scheme is 85 % efficient.

(a) Calculate the gravitational potential energy lost by the water each second. [2]

(a) Answer: _____ W

(b) Calculate the electrical power output. [1]

(b) Answer: _____ kW

(c) Calculate the electrical energy produced in one day, in kWh. [1]

(c) Answer: _____ kWh

(d) Calculate the maximum possible speed of the water at the turbine. [2]

(d) Answer: _____ m s^{-1}

(e) Suggest two reasons why the scheme is not 100 % efficient. [2]

3.4 Bouncing ball

A.3 · Energy conservation · [7 marks]

A 57 g tennis ball is dropped from rest from a height of 1.20 m. It rebounds to a height of 0.84 m. Air resistance is negligible.

(a) Calculate the speed of the ball just before it hits the floor. [1]

(a) Answer: _____ m s^{-1}

(b) Calculate the speed of the ball just after it leaves the floor. [1]

(b) Answer: _____ m s^{-1}

(c) Calculate the energy lost in the bounce. [2]

(c) Answer: _____ J

(d) Calculate the efficiency of the bounce. [1]

(d) Answer: _____ %

(e) Assuming the same efficiency each time, calculate the height reached after the third bounce. [2]

(e) Answer: _____ m

3.5 Ballistic pendulum A.3 · Energy conservation · [8 marks]

A 8 g bullet is fired horizontally into a 1.2 kg wooden block hanging on light strings. The bullet stays in the block, which swings up through a vertical height of 4.0 cm.

(a) Calculate the speed of the block (with the bullet) just after the impact. [2]

(a) Answer: _____ m s^{-1}

(b) Calculate the speed of the bullet before the impact. [2]

(b) Answer: _____ m s^{-1}

(c) Calculate the percentage of the bullet's kinetic energy that is lost in the impact. [3]

(c) Answer: _____ %

(d) Explain why energy conservation cannot be used to link the bullet's speed directly to the height of the swing. [1]

3.6 E-bike range A.3 · Power and efficiency · [7 marks]

An e-bike travels at a constant 7.0 m s^{-1} against total resistive forces of 28 N, using only its motor. The battery stores 0.60 kWh and the motor is 85 % efficient.

(a) Calculate the useful power of the motor. [1]

(a) Answer: _____ W

(b) Calculate the energy stored in the battery in joules. [1]

(b) Answer: _____ J

(c) Calculate the time for which the battery can drive the motor. [2]

(c) Answer: _____ s

(d) Calculate the range of the e-bike in km. [1]

(d) Answer: _____ km

(e) Explain why the range is shorter when the rider goes faster. [2]

3.7 Car climbing a hill A.3 · Power and efficiency · [7 marks]

A 1200 kg car climbs a hill inclined at 5.0° at a constant speed of 20 m s^{-1}. The total resistive force on the car is 600 N.

(a) Calculate the driving force needed. [2]

(a) Answer: _____ N

(b) Calculate the useful power output of the engine. [1]

(b) Answer: _____ kW

(c) The engine is 30 % efficient and petrol has an energy density of 46 MJ kg^{-1}. Calculate the mass of petrol used per kilometre of the climb, in grams. [3]

(c) Answer: _____ g

(d) State where the useful energy goes during the climb. [1]

3.8 Bungee jump A.3 · Energy conservation · [8 marks]

A 70 kg bungee jumper steps off a platform. The cord has an unstretched length of 18 m and behaves like a spring with k = 100 N m^{-1}. Air resistance is negligible.

(a) Calculate the speed of the jumper when the cord just becomes tight. [1]

(a) Answer: _____ m s^{-1}

(b) Calculate the extension of the cord when the jumper finally hangs at rest. [1]

(b) Answer: _____ m

(c) Show that the maximum extension of the cord is about 24 m. [3]

(c) Answer: _____ m

(d) Calculate the maximum force exerted by the cord on the jumper. [1]

(d) Answer: _____ N

(e) Calculate the maximum acceleration of the jumper as a multiple of g. [2]

(e) Answer: _____

End of exercises